

ARC 225 - Land Development

Project - Phase 1

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PROPERTY #: 01

The calculation for the total expenses for Property 1:

- Feasibility: $\$1/\text{SF} * 6944.44\text{SF} = \6944.44
- Land Acquisition: $\$0.50/\text{SF} * 6944.44\text{SF} = \3472.22
- Construction Loan: $\$0.25/\text{SF} * 6944.44\text{SF} = \1736.11
- Taxes and Fees: $\$2/\text{SF} * 6944.44\text{SF} = \13888.88

Total Expenses: $\$6944.44 + \$3472.22 + \$1736.11 + \$13888.88 = \$26,040.65$

Therefore, the total expenses for Property 1 is \$26,040.65.

Here is the calculation for the revenue for Property 1:

Revenue: $\$7/\text{SF} * 6944.44\text{SF} = \$48,611.08$

Therefore, the total revenue for Property 1 is \$48,611.08.

Here is a summary of the total expenses and total revenue for Property 1:

<u>Expense</u>	<u>Amount</u>
Feasibility	\$6944.44
Land Acquisition	\$3472.22
Construction Loan	\$1736.11
Taxes and Fees	\$13888.88
Total Expenses	\$26040.65
Revenue	\$48611.08
Net Profit	\$22569.43

Therefore, **Property 1 has a net profit of \$22569.43.**

The net profit margin for Property 1 is:

Net profit margin = net profit/revenue = $22569.43 / 48611.08 = 46.4\%$

The net profit margin for Property 1 is 46.4%. This means that for every \$100 in revenue, Property 1 generates \$46.40 in net profit.

Property 2 has a higher net profit margin of 64.9%, so it is more profitable than Property 1.

PROPERTY #: 02

The calculation for the total expenses for Property 2:

- Feasibility: $\$1.25/\text{SF} * 6944.44\text{SF} = \8680.55
- Land Acquisition: $\$0.75/\text{SF} * 6944.44\text{SF} = \5186.19
- Construction Loan: $\$0.50/\text{SF} * 6944.44\text{SF} = \3472.22
- Taxes and Fees: $\$2.50/\text{SF} * 6944.44\text{SF} = \1736.11

Total Expenses: $\$8680.55 + \$5186.19 + \$3472.22 + \$1736.11 = \$22075.07$

Therefore, the total expenses for Property 2 is \$22,075.07.

Revenue: $\$9/\text{SF} * 6944.44\text{SF} = \$62,493.36$

Therefore, the total revenue for Property 2 is \$62,493.36.

Here is a summary of the total expenses and total revenue for Property 2:

Expense	Amount
Feasibility	\$8680.55
Land Acquisition	\$5186.19
Construction Loan	\$3472.22
Taxes and Fees	\$1736.11
Total Expenses	\$22075.07
Revenue	\$62493.36
Net Profit	\$40418.29

Therefore, **Property 2 has a net profit of \$40418.29.**

The net profit margin for Property 2 is:

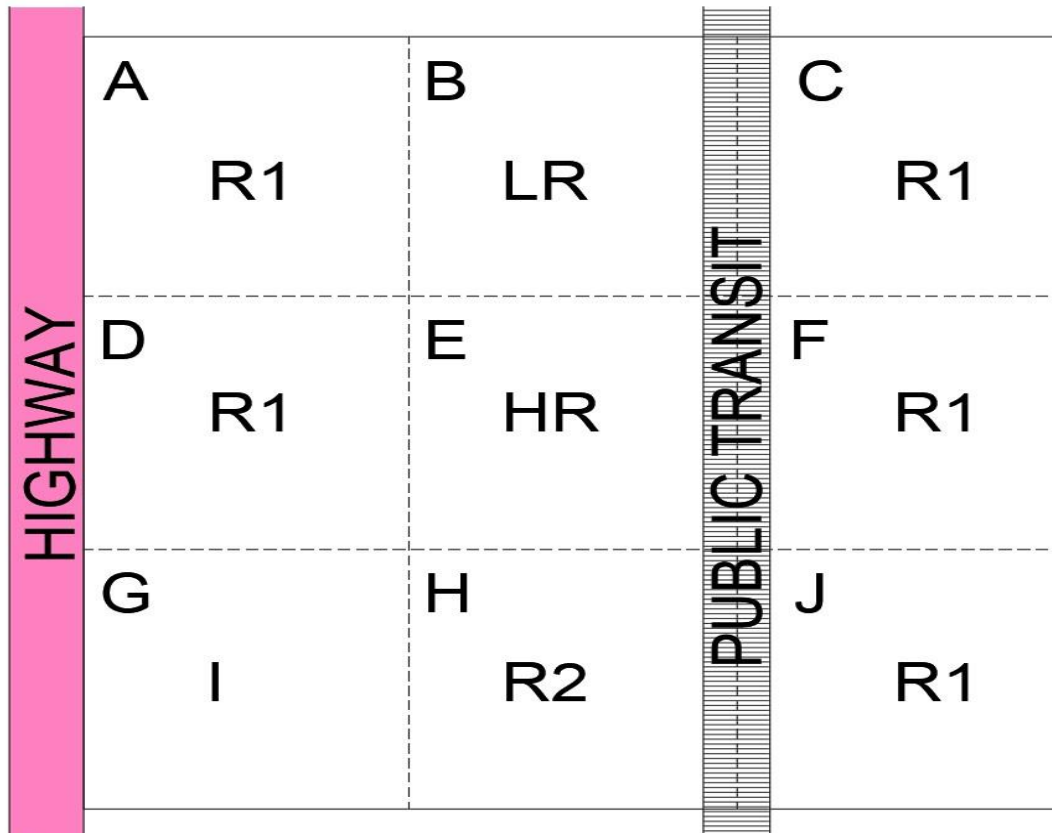
Net profit margin = net profit / revenue = $40418.29 / 62493.36 = 64.9\%$

The net profit margin for Property 2 is 64.9%. This means that for every \$100 in revenue, Property 2 generates \$64.90 in net profit.

Therefore, Property 2 is more profitable than Property 1.

Project - Phase 2

ZONING PRACTICE



NOTE:

It satisfies all of the guidelines and it also takes into account the needs of the different land uses. For example, the HR is located near public transit in order to attract residents and businesses. The R2 and I are located near each other in order to collaborate effectively. And the R1 and LR can be located in any zone, which gives them flexibility.

Overall, this is a well-thought-out arrangement that is both functional and attractive.

Project - Phase 3

QUESTION #: 01 2 FAR (in SF) with 50 % max site coverage (in SF).

FAR = Floor Area Ratio = Building Floor Area / Site Area

Maximum site coverage = 50%

Site area = 1000'x1000' = 1,000,000 square feet

Maximum building area = 50% * 1,000,000 square feet = 500,000 square feet

Maximum building volume = FAR * Maximum building area = 2.0 * 500,000 square feet = 1,000,000 cubic feet

Therefore, the maximum building volume on a 1000'x1000' site with a FAR of 2.0 and 50% maximum site coverage is 1,000,000 cubic feet.

QUESTION #: 50% max. Residential use. (Percentage) & (SF).

50% max. Residential use on a 1,000'x1,000' site

Percentage: 50%

Square feet: 500,000 SF (50% * 1,000,000 SF)

This means that up to 500,000 square feet of the site can be used for residential purposes. This could include apartments, condominiums, or houses.

QUESTION #: 50% max. Office use. (Percentage) & (SF).

50% max. office use on a 1,000'x1,000' site

Percentage: 50%

Square feet: 500,000 SF (50% * 1,000,000 SF)

This means that up to 500,000 square feet of the site can be used for office purposes. This could include a single office building, a mixed-use building with office space, or a combination of both.

QUESTION #: 10% min. retail use. (Percentage) & (SF).

10% min. retail use on a 1,000'x1,000' site

Percentage: 10%

Square feet: 100,000 SF (10% * 1,000,000 SF)

This means that at least 100,000 square feet of the site must be used for retail purposes. This could include a shopping mall, a strip mall, or a mix of individual retail stores. The remaining 900,000 square feet of the site could be used for other purposes, such as residential, office, or open space.

Project - Phase 4

QUESTION #: Site 100'x100', depth 20' - Calculate cubic feet

To calculate the volume, I use the following formula:

Volume = Length * Width * Depth

In this case, the length is 100 feet, the width is 100 feet, and the depth is 20 feet. Therefore, the volume is:

Volume = 100 feet * 100 feet * 20 feet = 200,000 cubic feet

This means that you will need to remove 200,000 cubic feet of dirt or other material from the site in order to excavate to a depth of 20 feet.

QUESTION #: Site 100'x100' , depth 20'

Calculate the Excavation cost knowing that the cost per CF = \$70.

200,000 cubic feet x 70\$ per CF = 14,000,000

QUESTION #: Site 100'x100' , depth 20'

Calculate the Fill cost knowing that the cost per CF = \$50.

200,000 cubic feet x 50\$ per CF = 10,000,000

0.50 re-fill cost

Find out if there will be hauling out excavated soil or bringing more soil for fill.

For a 100' x 100' site with a depth of 20', the volume of soil that needs to be moved is 200,000 cubic feet. This is a significant amount of soil, and it is important to have a qualified engineer or surveyor assess the site conditions and determine the best way to move the soil.

QUESTION #: Hauling the Excavated soil out of site cost per CF = \$10

To haul out excavated soil from a 100' x 100' site with a depth of 20' and a cost of \$10 per cubic foot, the total cost would be \$200,000. This is calculated by multiplying the volume of soil (200,000 cubic feet) by the cost per cubic foot (\$10).

Calculation:

Cost = Volume of soil * Cost per cubic foot

Cost = 200,000 cubic feet * \$10 per cubic foot

Cost = \$200,000

It is important to note that this is just an estimate of the cost. The actual cost may vary depending on a number of factors, such as the type of soil, the distance to the disposal site, and the availability of labor and equipment.

QUESTION #: Dumping the Excavated soil out of site cost per CF = \$5.

To dump the excavated soil out of a 100' x 100' site with a depth of 20' and a cost of \$5 per cubic foot, the total cost would be \$100,000. This is calculated by multiplying the volume of soil (200,000 cubic feet) by the cost per cubic foot (\$5).

Calculation:

Cost = Volume of soil * Cost per cubic foot
Cost = 200,000 cubic feet * \$5 per cubic foot
Cost = \$100,000

QUESTION #: Hauling soil from outside the site cost per CF = \$5.

It is important to note that this is just an estimate of the cost. The actual cost may vary depending on a number of factors, such as the type of soil, the distance to the disposal site, and the availability of labor and equipment.

To haul in soil from outside the site to a 100' x 100' site with a depth of 10' and a cost of \$5 per cubic foot, the total cost would be \$50,000. This is calculated by multiplying the volume of soil (100,000 cubic feet) by the cost per cubic foot (\$5).

Calculation:

Cost = Volume of soil * Cost per cubic foot
Cost = 100,000 cubic feet * \$5 per cubic foot
Cost = \$50,000

It is important to note that this is just an estimate of the cost. The actual cost may vary depending on a number of factors, such as the type of soil, the distance to the source of the soil, and the availability of labor and equipment.